

BYOTEE Extension Plan: Octavo

1 Required Software and Helpful Links

For hardware development, we will use **Vivado Design Suite**, Version 2024.1 or later. The software is available for Linux and Windows Os. Installation images can be found in the following link: <https://www.xilinx.com/support/download.html>

For software development, we will use **Vitis IDE** with the same version as the Vivado version. Installation images are in the following link: <https://www.xilinx.com/support/download/index.html/content/xilinx/en/downloadNav/vitis.html>

The development board documentation: https://octavosystems.com/octavo_products/osdzu3/

2 Extension Steps

2.1 Development Process

To become familiar with the hardware and software development process, we will start by developing a toy example using **Zynq** as the processor, along with push buttons and LEDs as **AXI GPIO** peripherals in the RTL design for the **OSDZU3-REF** development platform. The hardware design example for the project is shown in Figure 1.

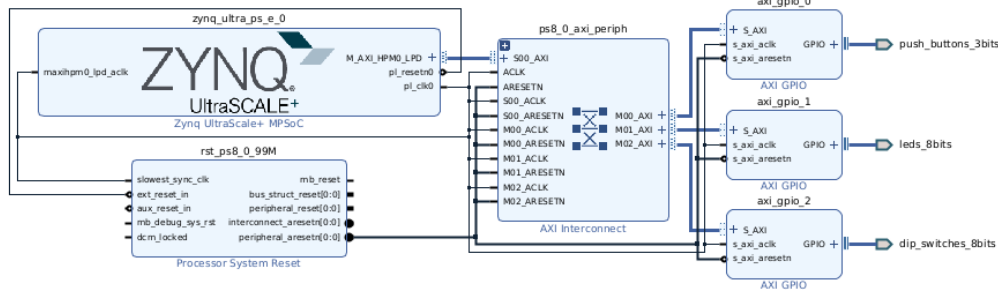


Figure 1: Hardware design for the simple application.

The **OSDZU3** development board, BSP installation, Vivado step-by-step descriptions for IP usage, Synthesis, Implementation, and Bitstream generation can be found in the following link: https://octavosystems.com/app_notes/osdzu3-vivado-tutorial/#Creating%20HDL%20Wrapper%20File%20for%20the%20Design

We can also extend this hardware design by adding MDM IP that will enable debugging over JTAG in vivado. Please refer to the Figure 4 to see an example hardware design.

Octavo also provides the step by step instructions for a simple software development process in the following link: https://octavosystems.com/app_notes/osdzu3-vivado-tutorial/#Creating%20HDL%20Wrapper%20File%20for%20the%20Design. Example out of the developed application is shown in Figure 2

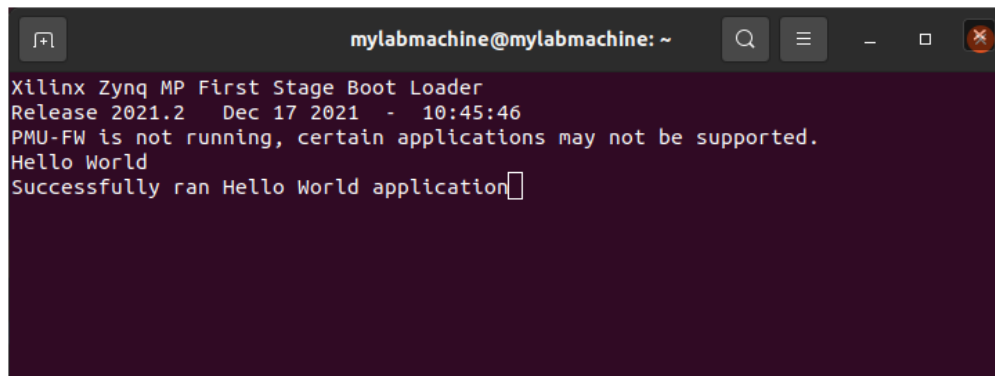
A terminal window titled 'mylabmachine@mylabmachine: ~' with a dark purple background. The text inside the terminal is white and reads: 'Xilinx Zynq MP First Stage Boot Loader', 'Release 2021.2 Dec 17 2021 - 10:45:46', 'PMU-FW is not running, certain applications may not be supported.', 'Hello World', and 'Successfully ran Hello World application'.

Figure 2: Output of the simple application.

2.2 RISC-V on the Programming Logic

AMD provides a RISC-V soft-core IP named [MicroBlaze V](#) off-the-shelf available on Vivado IDE. In this task, we will develop a simple example application to run a “Hello World” application. Both hardware and software development are required. Extend the hardware design in Figure 1 to add [MicroBlaze V](#) IP with its separate memory BRAM. The goal of this step test the RISC-V and get familiar with the software development process for PL.

Helpful documents:

- Vivado Design Suite Tutorial Embedded Processor Hardware Design
- <https://docs.amd.com/r/en-US/ug1579-microblaze-embedded-design/Using-a-MicroBlaze-F>
- BYOTEE repository: <https://github.com/CactiLab/BYOTee-Build-Your-Own-TEEs/tree/master/app1/pl>
- https://octavosystems.com/app_notes/osdzu3-vivado-tutorial/

2.3 PetaLinux Setup for Zynq and Shared Memory

Next task is to setup Peta Linux on the PS side. As highlighted in Figure 3, we are developing the software modules on top of the Cortex-A processor.

We can use/extend the simple hardware in Figure 1 or the hardware developed in the previous step.

Goals of this step:

- Get Peta Linux running on the PS
- Octavo provides the BSP file: https://octavosystems.com/octavo_products/osdzu3-ref/
- Share the part of the memory as UIO device
- Add GPIO interrupt for communication.
- Develop untrusted applications to execute on top of PetaLinux
- Test shared memory access from MicroBlaze and Zynq

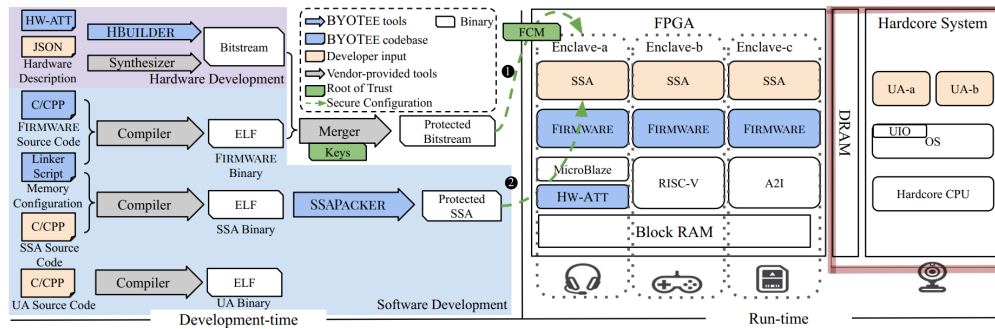


Figure 3: Processing System Development.

The Zynq and Microblaze shared memory access testing code can be found in the BYOTEE repository.

Zynq Untrusted Application: https://github.com/CactiLab/BYOTee-Build-Your-Own-TEEs/blob/master/app1/Untrusted_app/src/main.c

MicroBlaze Code: https://github.com/CactiLab/BYOTee-Build-Your-Own-TEEs/blob/master/app1/mb/AES_SSC/src/main.c

2.4 Target Hardware Setup

The target hardware will be very similar to the ones from the BYOTEE paper, with one or multiple Soft Core IPs, with example application development. Figure 4 shows an example from the paper with one soft core IP, Zynq, peripherals, and MDM for debugging them.

For each application, please refer to the hardware design of the SSA in the BYOTEE paper.

2.5 Target Software Setup

We will start with the applications that were used in the BYOTEE project. There are four SSAs in the paper. Below, are the short descriptions of the applications:

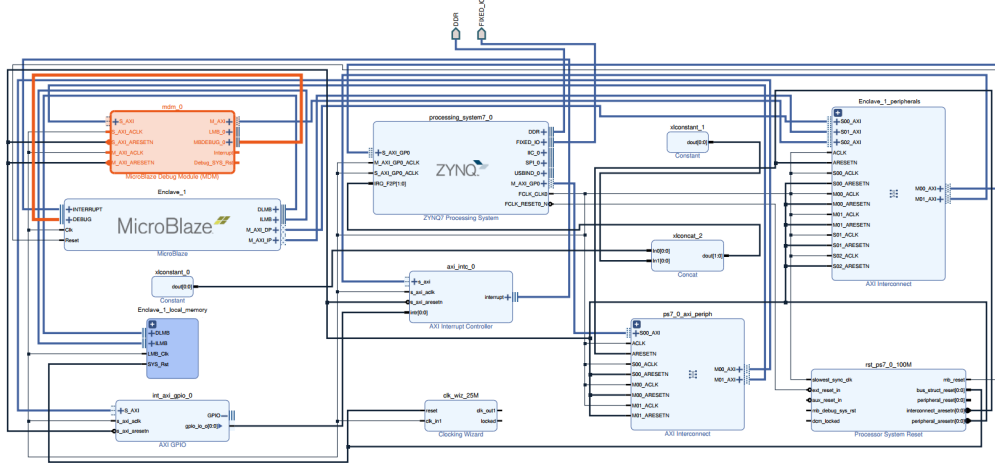


Figure 4: Target Hardware Setup.

- **SSA 1: AES accelerator-** The goal of this application is to execute the encryption algorithm on top of the softcore securely.
- **SSA 2: LED toggler-** In this application, the LED and button peripherals are dedicatedly controlled by the PL (softcore). It demonstrated the security benefits of the project.
- **SSA 3: DRAM music player-** This application shows the capability of the BY-OTEE design to ensure security properties of the DRAM application. Note: this application requires a **PMOD** module connected over I²C bus to play the audio data. The OSDZU3-REF development platform comes with an I²C bus, but we need to check if this application is suitable for this board.
- **SSA 4: Distributed application-** This application required two SSAs on the PL side, one for data decryption and the other to take the output of the first SSA and perform a digital signature verification. Communication between the two SSAs occurs over the shared BRAM memory.

ML inference application options:

- **LiteRT** is designed to run ML models for microcontrollers, developed in C++. Requires 16KB of memory, does not require OS support, dynamic memory allocation, or libraries. It's supported for RISC-V CPUs and is open sourced: [github](#)
- **MCUNet**
- **Tiny ML**
- ...

Note: DSP in the PL can accelerate ML models

3 Time Line

- **2.1:** 1-2 weeks
- **2.2:** 2-3 weeks
- **2.3:** 3-4 weeks
- **2.4:** 3 weeks
- **2.5:** 3 weeks

4 Notes from Octavo Meeting

- Got the two boards. The boards are OK. There is documentation on the website.
- We need 2 weeks to get used to the boards.
- Applications for edge devices (AI applications on top of Softcore)
- Data stored on edge device, and we carry out AI processing in the edge system
- Some AI examples for APPs.
- Figure out a scenario where local data processing in edge systems is useful for security (i.e., privacy). The AI model owner can share the model, but the data stays on the edge
- Edge device. Use data at the edge and combine AI. The data will not go to the cloud. However, the refined model could be shared with other AI owners. Federated learning may be here.
- The current board does not have a GPU. Let us use CPU or softcore CPU to do the AI.
- RISC-V softcore instead of MicroBlaze
- Do not open-source it now. But eventually we will.
- Bi-weekly meeting.